

CONCISE CLINICAL REVIEW

Electrical Impedance Tomography to Monitor Hypoxemic Respiratory Failure

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Abstract

Hypoxemic respiratory failure is one of the leading causes of mortality in intensive care. Frequent assessment of individual physiological characteristics and delivery of personalized mechanical ventilation (MV) settings is a constant challenge for clinicians caring for these patients. Electrical impedance tomography (EIT) is a radiation-free bedside monitoring device that is able to assess regional lung ventilation and changes in aeration. With real-time tomographic functional images of the lungs obtained through a thoracic belt, clinicians can visualize and estimate the distribution of ventilation at different ventilation settings or following procedures such as prone positioning. Several studies have evaluated the performance of EIT to monitor the effects of different MV settings in patients with acute respiratory distress syndrome, allowing more personalized MV. For instance, EIT could help clinicians find the positive end-expiratory

pressure that represents a compromise between recruitment and overdistension and assess the effect of prone positioning on ventilation distribution. The clinical impact of the personalization of MV remains to be explored. Despite inherent limitations such as limited spatial resolution, EIT also offers a unique noninvasive bedside assessment of regional ventilation changes in the ICU. This technology offers the possibility of a continuous, operator-free diagnosis and real-time detection of common problems during MV. This review provides an overview of the functioning of EIT, its main indices, and its performance in monitoring patients with acute respiratory failure. Future perspectives for use in intensive care are also addressed.

Keywords: hypoxemic respiratory failure; electrical impedance tomography; mechanical ventilation monitoring; ventilator-induced lung injury; acute respiratory distress syndrome

Hypoxemic respiratory failure requiring mechanical ventilation (MV) is a frequent cause of admission to ICUs, with acute respiratory distress syndrome (ARDS)

representing 10% of all admissions (1). Protective MV is the cornerstone for the management of these patients to limit ventilator-induced lung injury (2). However,

assessing the impact of MV on regional lung mechanics remains a daily challenge for ICU clinicians. The lung lesions in ARDS are heterogeneous and heavily influenced by

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gravity, and the regional distribution of air is mostly unpredictable (e.g., interstitial or alveolar lesions, unilateral pneumonia).

Advanced imaging techniques that are able to assess the regional impact of MV have been developed to enhance the personalization of protective ventilation. During a period of three decades, chest electrical impedance tomography (EIT) was developed for critically ill patients to provide relevant continuous, real-time, noninvasive, radiation-free bedside imaging of regional lung aeration and ventilation (3). EIT is now available in the clinical setting, and its application could highlight the pathophysiological processes occurring during hypoxemic respiratory failure and be used to monitor the consequences of MV on the different parts of the lungs. To guide its clinical implementation and future research questions, this review will provide an overview of the function of EIT and describe the information provided by this technique, its potential utility in daily clinical practice for the management of patients with hypoxemic respiratory failure, and its future prospects for use in the ICU.

Physical Principles of EIT

Technical Aspects

The physical principles and function of EIT have been described elsewhere (3, 4). Briefly, an electric current is applied through a thoracic silicone belt placed between the fourth and fifth intercostal spaces and surrounded by electrodes. By measuring the resulting voltage changes induced by the variation of air content (i.e., impedance change), EIT provides a bidimensional lung image through complex reconstruction algorithms (5) (Figure 1). Of note, external sources of electric current may interfere with EIT measurements, such as impedance pneumography and electrocautery. In addition, changes in body position, mattress movements, and spontaneous or passive movements of the patient can also modify the images (6). When long-term measurements are required, a standardized approach to keep the patient in the same position is important. Although variation in intrathoracic impedance is mostly due to changes in air content, cardiovascular impedance changes

also affect the measured signal, but this component is often removed from the signal (or used for perfusion measurements). Furthermore, a rapid variation in intrathoracic volume (e.g., a fluid challenge) could also affect measurements (7).

All electrodes should be in direct contact with the skin, avoiding interposed bandages. Significant pneumomediastinum or subcutaneous emphysema will drastically alter the electrical signal conduction and require careful interpretation. With proper interpretation, EIT can be used as a very sensitive method to detect small pneumothoraces (8).

Information Provided by EIT Devices and EIT-derived Indices

The EIT monitor typically presents the summed variation of impedance values (i.e., sum of pixel values) over time as a curve of “lung volume” versus time (the so-called

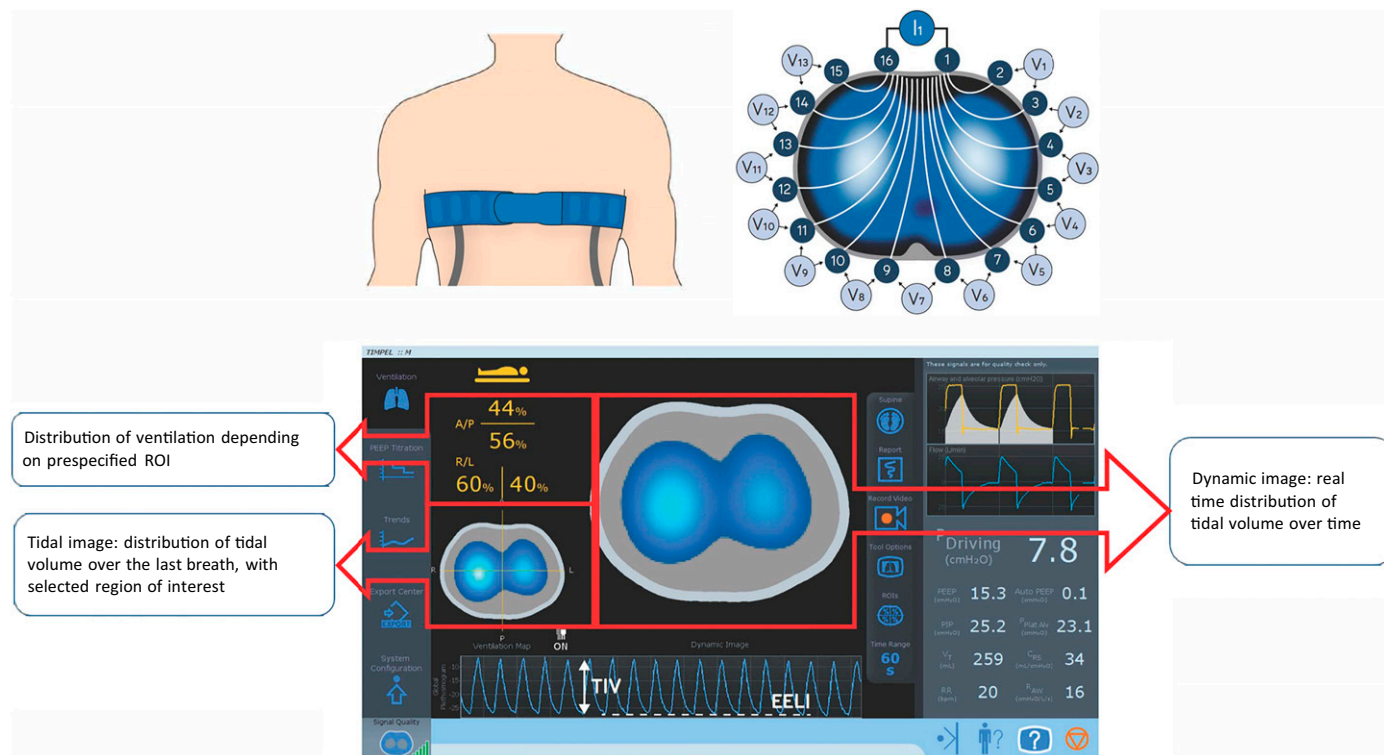


Figure 1. Principles of thoracic electrical impedance tomography. A belt composed of 16–32 electrodes is positioned between the fourth and fifth intercostal spaces. Each pair of consecutive electrodes delivers an electric current. The variation of the impedance of the crossed tissues is captured by other electrodes during a rapid rotation around the thorax. Thus, the variation of lung impedance induced by ventilation is converted into a bidimensional lung image. The dynamic image is a real-time representation of the impedance values of each pixel. The tidal image summarizes the tidal impedance variation during the previous breath. EELI = end-expiratory lung impedance; PEEP = positive end-expiratory pressure; ROI = region of interest; TIV = tidal impedance variation.

EIT plethysmogram) based on the proven concept that there is a linear relationship between regional impedance changes and regional changes in air content (9, 10). The tomographic representation of the lungs displays the regional distribution of V_T for a given breath, the so-called ventilation map (Figure 1).

Cyclic Variation in Lung Volume

Because impedance values and aeration are very well correlated, the amplitude of global (or regional) tidal impedance variation (TIV) is proportional to global (or regional) volume variation of the lungs during tidal ventilation. Variation of impedance can be calculated for the whole lung, a region of interest (ROI), or each pixel. Visual assessment of V_T distribution allows the identification of regions with unexpected decreases or absences of ventilation (e.g., lung collapse, pneumothorax, or pleural effusion) (9, 11).

Variation of End-Expiratory Lung Volume

Variations of end-expiratory lung impedance (EELI) could theoretically be used to assess variations of end-expiratory lung volume (EELV) caused by, for example, changes in ventilator settings or clinical evolution (12). Body positioning can also change EELI in a global or localized fashion (13). When the infusion of fluids is constant and a rather neutral fluid balance during the day is maintained, the changes in EELI correlate with the changes in EELV measured by whole-lung computed tomography (CT) with good precision (10). However, prior research employing various comparative methods, such as nitrogen wash-in/wash-out and helium dilution, indicated that the association between EELI and EELV may be weaker (14). Finally, fast boluses of saline solution or electrolytes can cause artifactual decreases in plethysmography results and have to be avoided during long-term EIT monitoring (7, 10).

Additional Indices and Examples of Potential Clinical Application

Ventilation inhomogeneity. The global inhomogeneity index is based on the sum of the differences between the impedance variation values of each pixel (during a respiratory cycle) and the median value of cyclic impedance variation, representing the spread of ventilation distribution (15). This sum is then divided by the sum of all impedance variations of every pixel along a

respiratory cycle, so the inhomogeneity is normalized between each breathing cycle and between patients.

The ventilation distribution has also been described by simply calculating the ratio between dorsal V_T and global V_T , termed the dorsal fraction of ventilation (16). This ratio is obtained by dividing the EIT image into two opposite ROIs, each one representing 50% of the anteroposterior length; a ratio close to 0.5 indicates a well-balanced anteroposterior ventilation (17). Finally, the center of ventilation (CoV), which is a geometrical representation of the CoV distribution (in analogy with the mathematical concept of center of mass), is another option to report the ventrodorsal or right-to-left distribution of ventilation (18). The ratio of dorsal to global V_T or CoV could help to identify hypoventilated lung regions, such as during a positive end-expiratory pressure (PEEP) trial. A dorsal/global V_T ratio largely lower than 0.5 represents lung ventilation mainly distributed in the nondependent (anterior) regions. A dorsal/global V_T ratio increase after increasing the PEEP indicates better ventilation in the dependent region, whereas an excessive PEEP typically leads to a V_T ratio higher than 0.5–0.55 as a result of overdistension of the nondependent lung sections (16).

Assessment of regional mechanics. The impedance variation can be divided by the driving pressure globally or for any defined region, including at the pixel level. EIT can thus describe regional compliance gain or loss at the pixel level induced by different ventilatory settings or body position changes. It can also be used to generate a pressure/volume curve for each predefined region or each pixel and obtain regional respiratory system compliance (19). The calculations above assume that the chest wall elastance is homogenous around the lung surface (20). Collapsed, recruited, and/or overdistended regions can be visualized assuming that a gain in compliance with lower PEEP indicates previous overdistension whereas a loss in compliance with lower PEEP indicates the appearance of collapse. The identification of “dependent silent spaces” could be another method for the evaluation of derecruitment. Briefly, pixels with <10% of the maximal TIV of overall pixels located in dependent regions (i.e., below the CoV) are considered as dependent silent space and could be used to identify poorly ventilated areas. The variation of these silent spaces measured by EIT could dynamically assess recruitment (21). Most

studies exploring these physiological applications of EIT are focused on mechanically ventilated patients with controlled modes (22), but similar principles can be applied to assisted and spontaneous breathing (23, 24).

“Optimal” PEEP based on lung collapse and overdistension. Costa and colleagues analyzed the changes in pixel compliance during a PEEP titration maneuver. For each pixel, the compliance curve has an inverted U shape, with the possibility of identifying the PEEP value associated with maximal pixel compliance. Below this maximum compliance, the pixel is losing some alveolar units (i.e., collapse), and, above this maximum, this pixel is overdistended. By computing a weighted sum of all pixels, it is possible to compute the total amount of lung collapse and overdistension for each PEEP level, resulting in a computation that presents a good agreement with CT quantification (25). After analyzing those global values of overdistension and collapse at each PEEP level, it is possible to choose the “best” overall compromise (i.e., minimal difference) between global overdistension and collapse. For this analysis, clinicians need to perform a decremental PEEP trial while maintaining the remaining ventilator settings constant. The highest PEEP is used as the reference for the minimal collapse achievable assuming that the lung is maximally open at this PEEP level. On the contrary, minimal overdistension is assumed when all pixels are presenting a collapsing behavior (25). The PEEP of the crossing point between the curves (global curves of lung collapse and overdistension) coincides with slightly positive transpulmonary pressure, typically approximately +2 cm H_2O . This technique has been validated in experimental and clinical studies (26, 27) and is a potentially relevant tool to personalize PEEP settings.

Limitations of this method should be mentioned. First, the choice of this “crossing-point PEEP” assumes that overdistention and collapse contribute equally to lung injury, which is still unknown (28). Second, values of overdistension and collapse depend on the range of PEEP explored. Other methods for PEEP titration using EIT have been proposed (29–31). Besides resulting in slightly different PEEP choices, all these other methods lack validation against independent and quantitative methods such as CT or absolute esophageal pressure measurements. It is important to analyze

the EIT images across a few PEEP steps. The most commonly used EIT indices are summarized in Table 1.

One important physiological caveat about “lung homogeneity” is that the greatest homogeneity can be achieved when the whole lung is completely overdistended, including the dependent lung zones. Therefore, setting PEEP based solely on homogeneity or “best ventilation distribution” cannot be recommended.

Temporal heterogeneity of regional mechanics. Lung injury is characterized by alveolar collapse, edema, and/or regional surfactant depletion, leading to heterogeneity of regional mechanics within a breath cycle and a cyclic opening-and-closing phenomenon during tidal ventilation. The regional ventilation delay reflects the time between the start of inspiration and the start of the aeration of lung regions (detected by an increment in the pixel value of 10% or

40% of the maximal value of TIV) and was well correlated with regional alveolar (i.e., tidal) recruitment in a pig model of lung injury (32). The greater the delay, the greater the impairment of that lung region. A similar analysis during expiration could be performed during this phase of the breathing cycle to illustrate lung heterogeneity. The time required for passive expiration, reflected by the expiratory time constant, could differ depending on lung

Table 1. Definition, Clinical Application, and Limitations of the Most Commonly Used EIT Indices in Patients with Hypoxemic Respiratory Failure

EIT Indices	Definition	Potential Applications	Limits
TIV	Variation of impedance values between end of inspiration and end of expiration (amplitude of plethysmogram signal)	- Reflects global and regional relative distribution of V_T in lungs during ventilation - Contributes to identification of pleural effusion, pneumothorax, endotracheal tube malposition, pendelluft	- Cannot be used for detailed anatomic diagnosis - Only relative distribution of V_T
EELI	Regional lung impedance at end of expiration	- EELI changes are correlated with EELV variation - Reflects lung recruitment/derecruitment - Contributes to identification of pneumothorax	Affected by factors other than lung volume (inflatable mattress, fluid infusion)
Global inhomogeneity index	Sum of absolute differences between median TIV and each pixel TIV normalized to sum of each pixel TIV	Reflects heterogeneity of ventilation	- Does not reflect local distribution of TIV - Does not take into account overdistension, collapse, or any other pathological situation
Overdistension and collapse estimation	Estimation of relative local compliance loss presented as percentage of collapse and overdistension during decremental PEEP trial	- Selection of PEEP level that jointly minimizes collapse and overdistension - Applicable even on venovenous ECMO when V_T is very low	- Assumes that ΔP on ventilator is reliable surrogate of regional ΔP - Depends on highest and lowest values of PEEP applied during PEEP trial
Regional ventilation delay	Time between start of inspiration and aeration of lung regions	- Identify recruitable lung regions and cyclic opening/closing phenomenon - Could be used to identify best PEEP level with most homogenized tidal inflation	- Requires a slow inflation maneuver with constant flow - Depends on defined threshold - Does not detect not-recruitable or overdistended regions
Center of ventilation	Describes geometrical center of ventilation	Reflects global ventilation homogeneity to guide and monitor impact of mechanical ventilation management or adjunct therapies (i.e., prone positioning)	Reflects ventilation shifts but not their origin
Dorsal fraction of ventilation	Describes percentage of V_T distributed in dorsal region	Identification of adverse events (e.g., malpositioning of endotracheal tube or development of pneumothorax)	- Imprecise interpretation of ventilation distribution (ventilation on one side can be compensated by other side). - Setting PEEP aiming for >50% dorsal distribution could imply overdistension of whole lung

Definition of abbreviations: ΔP = driving pressure; ECMO = extracorporeal membrane oxygenation; EELI = end-expiratory lung impedance; EELV = end-expiratory lung volume; EIT = electrical impedance tomography; PEEP = positive end-expiratory lung pressure; TIV = tidal impedance variation.

region, mechanics, and injury, and could be calculated with EIT (33).

Lung perfusion assessment. Signals associated with heartbeat or lung perfusion can be analyzed independently and coupled with ventilation. A hypertonic saline solution bolus (acting as a contrast agent to enhance the blood flow signal) injected during a brief apnea in passive patients allows clinicians to analyze lung perfusion individually (34) and in a quantitative fashion. When coupled with EIT ventilation, it could provide a quantification of ventilation/perfusion mismatch and may help to assess the degree of hypoxic vasoconstriction (35). It has also been used to understand the physiological effects of prone positioning (36), PEEP changes (37), and pulmonary embolism (38). Although this technique may be promising, it remains a research tool under evaluation.

Contribution of EIT to the Monitoring of Hypoxemic Respiratory Failure

By providing continuous bedside monitoring during ventilation, EIT allows: 1) evaluation

of the physiological impact of therapeutic or diagnostic procedures (e.g., endotracheal suction, BAL), 2) personalization of ventilator settings to enhance the prevention of ventilator-induced lung injury, and 3) monitoring of the effects of PEEP titration, prone positioning, and/or protective ventilation strategies (3). Figure 2 illustrates the potential use of EIT to optimize the management of patients with acute respiratory failure. Despite the various applications of EIT, robust data suggesting a potential clinical benefit of daily management of MV settings with EIT are still under investigation.

Basic Monitoring with Real-Time Analysis at the Bedside

EIT can be used to detect the occurrence of spontaneous or procedure-induced pneumothorax. An increase in regional thoracic impedance associated with a decrease in regional ventilation on EIT yields a detection of pneumothorax with 100% sensitivity compared with CT images (8). In addition, EIT enables clinicians to monitor the correct placement of the endotracheal tube (39). Similarly, the regional impact of several ICU procedures such as endotracheal

suction or BAL could provide relevant information at the bedside. Endotracheal suctioning has yielded a heterogeneous volume loss in the anterior regions (40) in addition to some irreversible loss of regional EELI (40), whereas a decrease in TIV and regional compliance in the BAL territory has been observed (41). Interestingly, this BAL-induced regional derecruitment observed in EIT seemed more important in a population with nonsevere ARDS (42). Also, unilateral lung injury with the exclusion of one side from ventilation is an interesting condition in which EIT monitoring could be extremely helpful. Usually, this unique information helps to optimize therapy, providing important clues to indicate further bronchoscopy, bronchial clearance, or a change in patient positioning that can help with blood gases or lung re-aeration (43).

Hypoxemic respiratory failure with noninvasive oxygenation strategies. The mechanisms underlying oxygenation and the possible harmful consequences of noninvasive oxygenation methods, such as high-flow nasal oxygen, remain uncertain. EIT has offered multiple physiological insights concerning these aspects. EIT

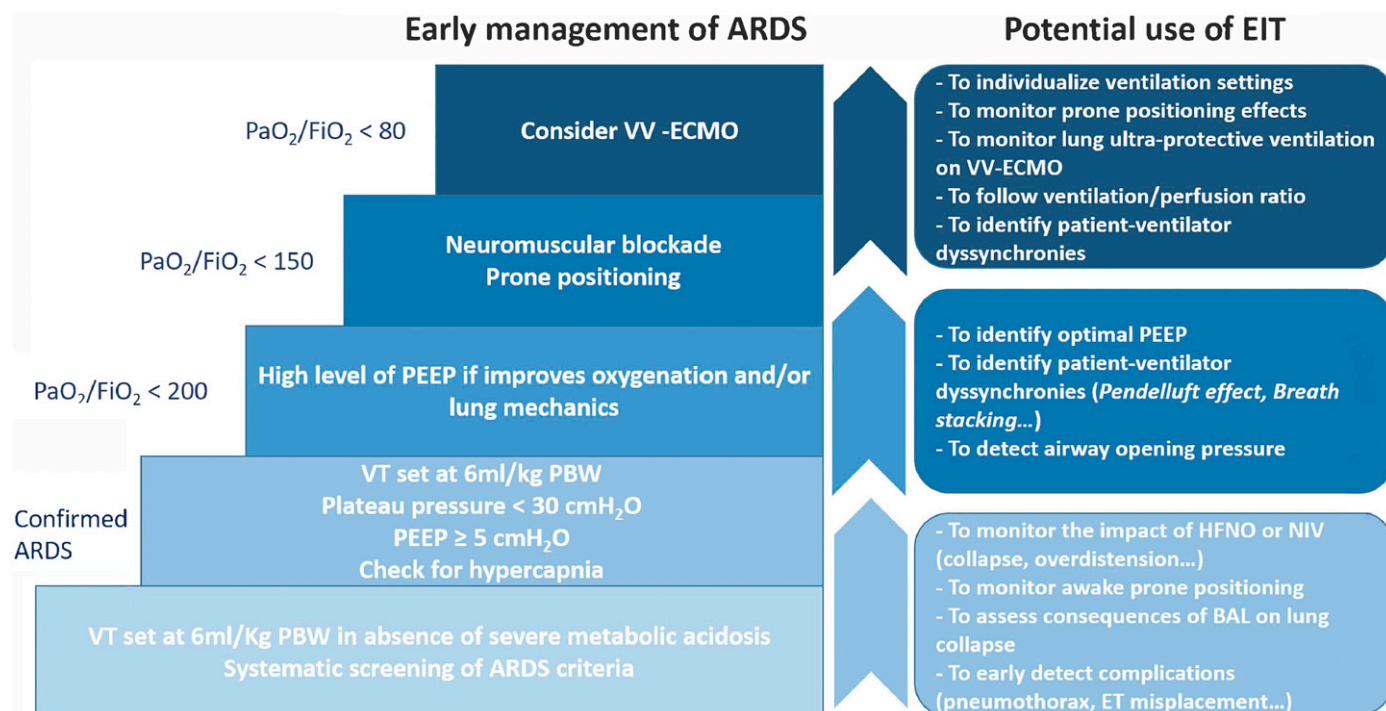


Figure 2. The potential use of electrical impedance tomography according to the severity of acute respiratory distress syndrome. The $\text{PaO}_2/\text{FiO}_2$ ratio is shown in mmHg. ARDS = acute respiratory distress syndrome; EIT = electrical impedance tomography; ET = endotracheal tube; HFNO = high-flow nasal oxygen; NIV = noninvasive ventilation; PBW = predicted body weight; PEEP = positive end-expiratory pressure; VV-ECMO = venovenous extracorporeal membrane oxygenation.

Table 2. Summary of Recent Clinical Trials Evaluating the Outcomes of an EIT-based PEEP in Patients with ARDS

Population	Severity of the Population	Mean PEEP in the EIT group	Mean PEEP in the control group	Results
24 patients with ARDS vs. 31 historical patients (26)	APACHE II score 23.5	Based on overdistension and collapse: 18 cm H ₂ O	2 cm H ₂ O above inflection point of pressure–volume curve: 14 cm H ₂ O	- Higher compliance in EIT-based PEEP group: 26 vs. 20 ml/cm H₂O - No difference in clinical outcomes
117 patients with ARDS (83)	19.7% severe ARDS, 45.3% moderate ARDS, APACHE II score 19	Based on overdistension and collapse: 8 cm H ₂ O	Based on PEEP/FiO ₂ table: 8 cm H ₂ O	- Nonsignificant 28-d mortality in EIT group: 21% vs. 27% - ICU length of stay: 13 vs. 10 d - Ventilator-free days: 14 vs 23 d
87 patients with ARDS (81)	75.9% severe ARDS, inclusion ≤24 h after intubation	Based on overdistension and collapse: 16.2 cm H ₂ O	Based on pressure–volume curve: 17.4 cm H ₂ O	- Lower hospital mortality in EIT group: 44.4% vs 69.0% - Lower ΔP in EIT group: 10.9 vs. 12.4 cm H₂O
12 patients with ARDS with a cross-over protocol (84)	PaO ₂ /FiO ₂ , 130; MV duration at inclusion, 0.6 d; SAPS II, 38	EIT-selected PEEP was lower: mean difference of change, –2 cm H ₂ O		- Lower mechanical power in EIT group: 11.42 J/min, with lower ΔP and higher static respiratory system compliance

Definition of abbreviations: APACHE = Acute Physiology and Chronic Health Evaluation; ARDS = acute respiratory distress syndrome; ΔP = driving pressure; EIT = electrical impedance tomography; MV = mechanical ventilation; PEEP = positive end-expiratory pressure; SAPS = Simplified Acute Physiology Score.

highlighted that high-flow nasal oxygen can increase EELV (illustrated by EELI) mainly in the dependent regions in a prone or supine position (44). The aeration of the dependent regions increases with the gas flow, but, as a counterpart, it could be associated with overdistension in the nondependent regions, as suggested by the decrease of TIV in these aerated regions (45).

PEEP selection. As a result of its regional and functional analysis, EIT is the only bedside monitoring tool that can be used to identify relative overdistension and collapse at the same time. Several trials have evaluated the performance of regional EIT-derived compliance to maximize the recruitment of dependent lungs and minimize overdistension of nondependent lung areas. These studies are summarized in Table 3. Briefly, human studies often performed a PEEP titration based on the estimation of overdistension and collapse, with conflicting results reported regarding the impact of an EIT-based PEEP on mortality. Different PEEP strategies in the

control arms, heterogeneous disease severity, or the variety of EIT-derived variables studied could explain these findings (Table 3). Notably, the optimal PEEP level could differ widely depending on the EIT indices selected (16, 27, 46, 47) (Figure 3). Ongoing randomized trials may help to assess the still unknown clinical benefits of an EIT-based PEEP strategy in patients ARDS (clinicaltrials.gov identifiers NCT04247477, NCT03793842, and NCT03112512).

Personalization of other ventilator settings and ventilation modes. A pilot feasibility study in 20 patients with ARDS investigated an EIT-based strategy to set V_T and PEEP depending on regional compliance and driving pressure (48). An increase in regional compliance after halving the driving pressure was interpreted as overdistension and led to a reduction of V_T. This EIT-based protocol led to individualized ventilator settings with improved oxygenation and lower values of regional ventilation delay. It was consistent with a reduction in alveolar cycling without causing excessive lung stress and strain (48). EIT enables the physiological

impacts of an open lung approach to be monitored (49, 50). In cases of respiratory failure with unilateral pneumonia, EIT can be used with two ROIs (left and right lungs) to find the best compromise between recruitment (i.e., PEEP) and overdistension (i.e., V_T) of each lung. The crossing point of the overdistension and collapse curves during a decremental EIT PEEP trial allows the PEEP to be set to recruit the sickest portion of the lungs and to reduce V_T in case of overdistension (51).

Personalization of various ventilatory modes has not been extensively studied so far. In 10 mechanically ventilated patients with acute lung injury, Neurally Adjusted Ventilatory Assist seemed to have a higher center-of-gravity index than pressure support ventilation, suggesting better ventilation of the dependent regions (52). This effect seems to be partially dependent on the PEEP level (53). Similarly, variable-pressure support ventilation led to more homogenous ventilation than pressure support alone (54).

Weaning from MV. Failure of a spontaneous breathing trial may be due

Table 3. Main Advantages and Limitations of EIT and Future Improvements to Address Limitations

Main Advantages	Main Limitations	Direct Impact on Clinical Practice	Potential Future Improvements
- Real-time monitoring of lung ventilation - Evaluation of regional compliance before and after therapeutic procedures (e.g., recruitment) - Visualization and quantification of overdistension and collapse - Identification of pathological situations (inhomogeneity, Pendelluft effect) - EIT can be coupled with ventilator for further physiological analysis - Noninvasive continuous bedside monitoring	- Low spatial resolution - Most sensitive to detect electrical impedance changes occurring in plane of electrode belt - Performance, relevance, and reference values of EIT indices still lacking - Lung perfusion assessment insufficiently developed - Unknown clinical benefit of daily EIT-guided mechanical ventilation strategy	- EIT cannot be used for precise anatomic diagnosis - EIT analysis cannot be extrapolated to lung part outside visualized area - EELV cannot be precisely inferred from EELI - Performance/relevance of available EIT indices to assess optimal PEEP have not been compared - Normal ranges of EIT indices unknown and may vary between patients - Most EIT devices still do not perform lung perfusion analysis - Evaluation currently requires apnea and hypertonic saline bolus - Clinical impacts of ventilation strategy guided by EIT remain unknown	- Technological improvement and/or increase in number of electrodes on belt - Further studies on potential benefit of combining apical EIT analysis with conventional one - Best EIT indices for each purpose need to be identified in patients with ARDS “Normal ranges” should be accurately evaluated - Integration of lung perfusion module in all EIT devices. - Promising method for lung perfusion without saline bolus should be further investigated - Few clinical trials are ongoing (NCT04247477, NCT03793842, NCT03112512) to assess potential benefit of EIT-guided mechanical ventilation on outcomes in patients with ARDS

Definition of abbreviations: ARDS = acute respiratory distress syndrome; EELI = end-expiratory lung impedance; EELV = end-expiratory lung volume; EIT = electrical impedance tomography; PEEP = positive end-expiratory pressure.

to lung derecruitment and ventilation inhomogeneity. Beyond the clinical signs of respiratory failure during a spontaneous breathing trial, EIT could directly quantify this derecruitment and may be used as a prognosis tool for weaning. Indeed, observational studies demonstrated that EIT can provide visualization and quantification of the spatial distribution of V_T during a spontaneous breathing trial and suggested that an increase in the global inhomogeneity index could predict the failure of this trial (55). Moreover, derecruitment (associated with a decrease in EELI) and greater inhomogeneity could also predict failure of the spontaneous breathing trial (56).

Advanced Monitoring and Need for Offline Analysis

Monitoring spontaneous breathing and detection of the Pendelluft phenomenon.

During spontaneous breathing, the decrease

in pleural pressure is greater in dependent regions, leading to a shift of volume from the anterior region to the posterior regions called the occult Pendelluft phenomenon (57, 58). It could contribute to patient self-inflicted lung injury because the impact of effort is not homogeneous and can have important regional effects (59). EIT allows its identification by different methods. The first one compares posterior and anterior ROIs (58), which could be considered imprecise because regions with and without Pendelluft could be part of the same ROI. Another method explored phase-shifts with a single-pixel analysis (60). However, this more sensible analysis requires significantly advanced offline treatment.

Breath stacking can result from reverse triggering or double triggering and may hinder lung-protective ventilation by potentially delivering greater volumes than intended (61), regional hyperinflation (with

increased histological patterns of ventilator-induced lung injury), and inefficient ventilation. In particular, reverse triggering was shown to cause a regional overdistension of dependent lung regions via the Pendelluft phenomenon (62).

Airway opening pressure detection.

Because of airway closure, tidal inflation starts only when an airway opening pressure (AOP) has been overcome. In at least one third of patients with ARDS, this AOP is >5 cm H_2O (63), leading to a complete collapse of the airways during expiration if PEEP is set too low. The detection of global airway closure and the measurement of a global (or minimal) AOP requires the performance of a low-flow insufflation maneuver, which can also be confirmed by EIT. Notably, another method to detect AOP without a low-flow insufflation maneuver was recently described (64). During a low-flow insufflation, EIT can detect gas

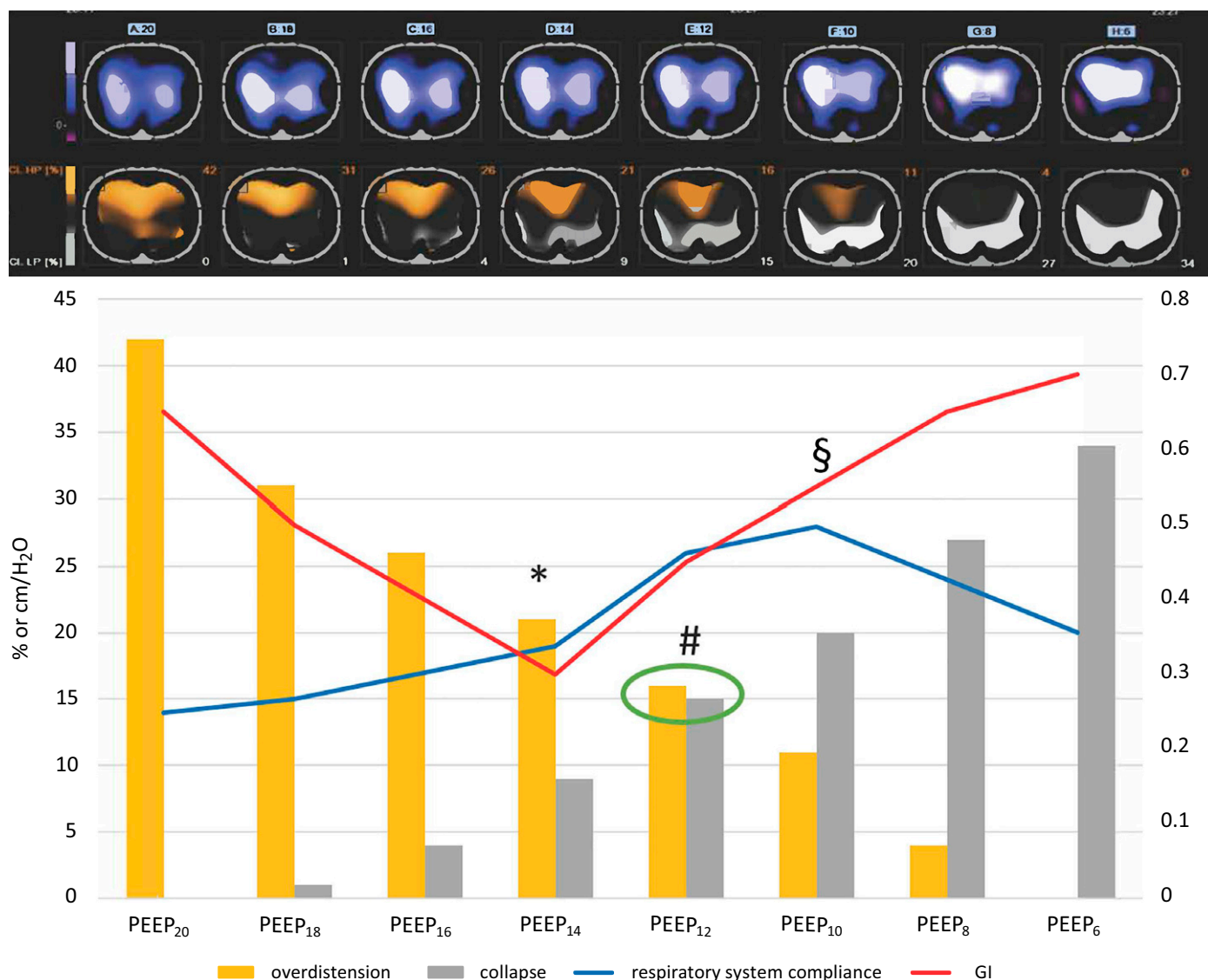


Figure 3. Positive end-expiratory pressure (PEEP) titration in a patient with acute respiratory distress syndrome with the corresponding estimation of the percentage of overdistension and collapse, global inhomogeneity (GI) index, and respiratory system compliance. The upper panel shows the tidal volume distribution at each PEEP level. The PEEP titration was performed over a few minutes and then analyzed. The lower line reports the distribution of the lung regions with a decrease in local compliance at high PEEP (i.e., overdistended lung; orange) and those with a decrease in local compliance at low PEEP (i.e., collapsed lung; white). The graph summarizes the percentage of overdistension and collapse, GI index, and respiratory system compliance at each PEEP level. Depending on the chosen method, for the same patient, the optimal PEEP could differ. Left y-axis: percentage of overdistension (yellow) and collapse (gray) and values of global compliance (ml/cm H₂O; blue line) at each PEEP level. Right y-axis: values of GI index (red line). *Optimal PEEP according to the lowest GI value: 14 cm H₂O. #Optimal PEEP according to the best compromise between overdistension and collapse: 12 cm H₂O (green circle). §Optimal PEEP according to the highest respiratory system compliance: 10 cm H₂O.

movement and its distribution by monitoring regional TIV, identifying the minimal pressure to open the airways globally or regionally (65). Additionally, the EIT-derived low-flow pressure–volume curve can identify differential airway closure in case of asymmetrical lung injury and allows the measurement of the respective AOP of each lung, whereas the global

pressure–volume curve on the ventilator shows mostly the behavior of the lung with the lowest AOP (51). In the context of asymmetrical ARDS, setting the PEEP based on the AOP obtained from the global pressure–volume curve (i.e., monocompartmental model) recorded by the ventilator may not allow the protection of the sickest lung from repeated opening and

closure (66). By analyzing the two lungs separately, EIT can identify different AOP levels between the lungs (Figure 4).

Prone and lateral positioning. Prone positioning leads to an early redistribution of V_T in the dependent regions and an increase in regional compliance when associated with an increase in PEEP (67). Similarly, prone positioning yielded a decrease in the global

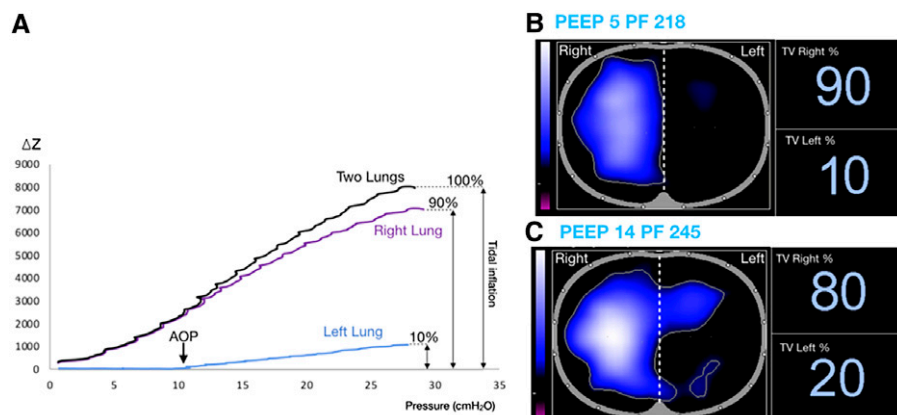


Figure 4. Electrical impedance tomography–derived pressure–volume curves in a patient undergoing invasive mechanical ventilation with unilateral left pneumonia. (A) Electrical impedance tomography–derived pressure–volume curves of both lungs and the right and left lungs individually in a patient with unilateral left pneumonia. This figure illustrates an airway opening pressure (AOP) of the left lung that is not captured when both lungs are analyzed simultaneously. (B) At a positive end-expiratory pressure of 5 cm H₂O (i.e., lower than the AOP of the left lung), tidal ventilation of the left lung represents only 10% of total tidal ventilation, with the PaO₂/FiO₂ ratio at 173 mm Hg. (C) At a positive end-expiratory pressure of 14 cm H₂O (i.e., higher than the AOP of the left lung), tidal ventilation of the left lung increases to 20% of total tidal ventilation, and the PaO₂/FiO₂ ratio increases to 245 mm Hg. PEEP = positive end-expiratory pressure; PF = PaO₂/FiO₂ ratio; TIV = tidal impedance variation.

inhomogeneity index without a significant change in lung perfusion (68). Thus, the ventilation–perfusion matching increases concomitantly with the PaO₂/FiO₂ ratio (69). It has been shown that lung recruitment induced by prone positioning monitored by EIT in patients with severe ARDS undergoing venovenous extracorporeal membrane oxygenation (ECMO) seems to decrease over sessions (70).

Few EIT-based studies have attempted to predict prone positioning response. In patients with coronavirus disease (COVID-19)–related ARDS, the proportion of dependent lung area collapse on supine positioning could predict oxygenation response after prone positioning (71). Moreover, EIT analysis highlighted the need to reevaluate PEEP levels in the prone position, with a high likelihood of a lower optimal PEEP in the prone position (70, 72) (Figure 5).

Lateral positioning in ventilated patients leads to similar consequences in regional ventilation as when PEEP is applied. In a cohort of 15 patients undergoing ventilation

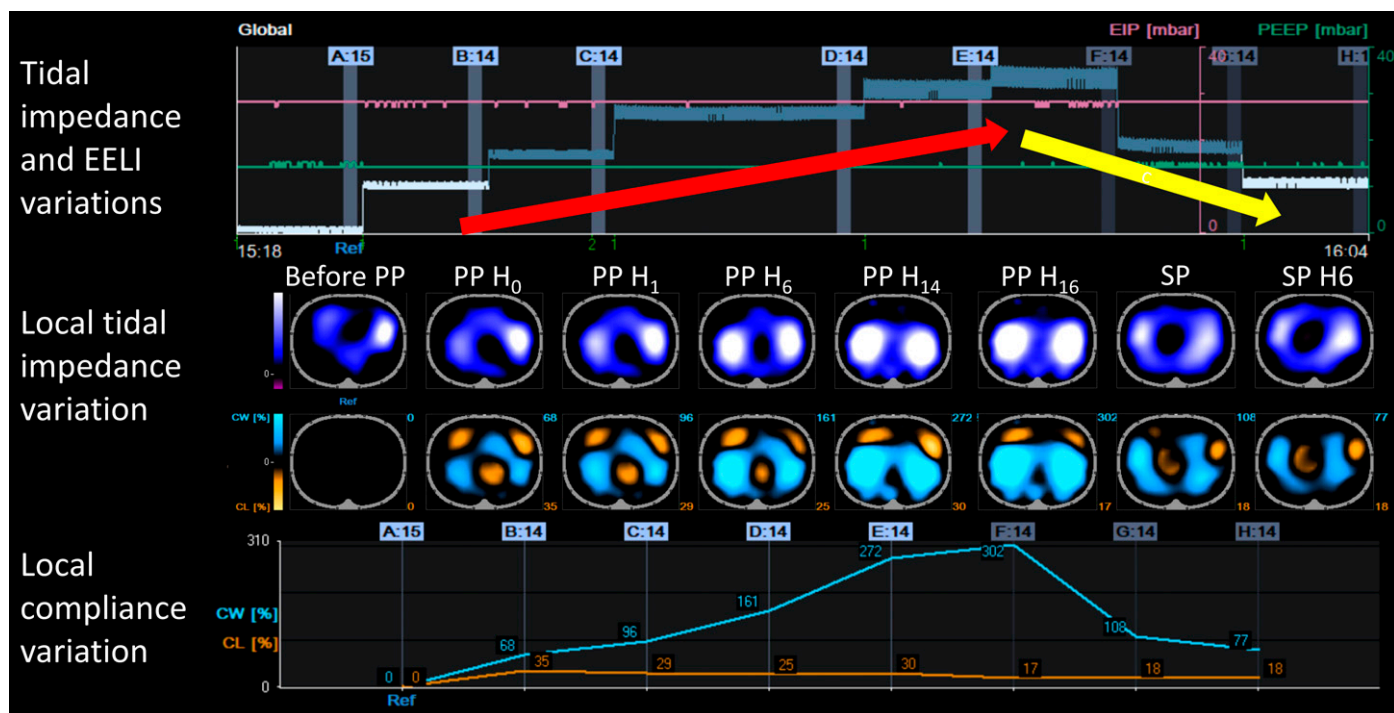


Figure 5. Prone positioning monitored by electrical impedance tomography. The upper graph shows impedance and end-expiratory lung impedance (EELI) variations before, during (16 h), and after prone positioning (at 6 h after back in supine position). The red arrow follows the increase of EELI associated with an increase in local tidal impedance variation (i.e., more white zones in the second line) and local compliance (i.e., more blue zones in the third line). The yellow arrow follows the decrease of EELI on supine positioning that is associated with derecruitment (i.e., decrease of local tidal impedance variation and local compliance compared with baseline [yellow zones]). EIP = electrical impedance pressure; PEEP = positive end-expiratory pressure; PP = prone positioning.

for COVID-19–related pneumonia, sequential lateral positioning (i.e., positioning the patient on each side sequentially) was found to cause an upward increase in EELI. Upon returning to the supine position, dorsal EELI was higher than at baseline, indicating recruitment and permanent improvement in aeration and regional compliance. This observation suggests a recruitment effect of sequential lateral positioning on lung mechanics and ventilation in these patients without increasing pressures (73). Of note, the results were less predictable and changed in patients breathing spontaneously (74).

Monitoring MV in patients receiving ECMO. In a study of 21 patients with the most severe form of ARDS treated with venovenous ECMO, prone positioning was associated with a redistribution of V_T during the 16-hour session, even for patients without gas exchange improvement (70). If V_T was distributed mainly into nondependent regions before prone positioning (suggesting a collapse of the dorsal regions), the patients' gas exchange was likely to improve after prone positioning. EIT monitoring could even be achieved during ECMO with very low V_T (46). During a decremental PEEP titration in 15 patients with ECMO, EIT enabled the personalization of ventilation settings and the identification of optimal PEEP based on overdistension and collapse estimation. This optimal PEEP was frequently different from the PEEP that led to the greatest respiratory system compliance (46).

Limitations of EIT

In the context of hypoxemic respiratory failure, several types of lung lesions (e.g., atelectasis, overdistension, lung fibrosis, and/or pleural effusion) could be observed in the same patient and visualized similarly as a lack of regional ventilation shown by EIT. Moreover, the percentage of collapse and overdistension may vary between lung regions outside the lung imaging field of EIT, which limits the analyses to a slice of the lungs. However, this slice is rather thick, and animal studies have suggested that the EIT slice best correlates with a thick CT slice of approximately 20 cm along the craniocaudal axis (75). Newer experimental studies performed with quantitative whole-lung CT, a technology with much greater precision than nitrogen washout, and great care

regarding fluid administration during measurements have shown extremely good agreement between changes in EELV and EELI (10). However, these results conflict with those of previous studies (14). The various techniques employed to measure EELV may account for these disparities.

Another limitation of EIT is the craniocaudal heterogeneity of the lung lesions in ARDS. In supine position and up to 15° versus the horizontal plane, this is not an important issue because of the liquid-like behavior of the lung (76), with EIT fully representing the gravity axis. However, this may not be true when the patient is elevated at a high angle (>30° vs. horizontal plane). Highlighting this specificity, Bikker and colleagues analyzed regional compliance and CoV during a PEEP trial by using two EIT belts positioned at two different levels of the thorax. These analyses showed different responses to the PEEP depending on the thorax level, with greater compliance achieved at a lower PEEP for nondependent regions (77). Moreover, EIT monitoring has a few contraindications, primarily due to the absence of studies demonstrating its safety in patients with pacemakers or implantable defibrillators.

EIT quantifies only the relative change in impedance. Consequently, a direct comparison of absolute values of impedance and/or EELI between patients is irrelevant. Similarly, the estimation of overdistension and collapse percentages during a PEEP trial will provide only relative values compared with the highest and lowest PEEP applied. In addition, some advanced EIT parameters (on a per-pixel basis) often require complex offline analyses that are not (yet) available at the bedside. Moreover, the use of EIT indices to guide MV settings needs to be standardized. The main advantages, limitations, and potential future improvements of EIT are summarized in Table 2.

Future Perspectives of EIT in Intensive Care

For two decades, many technical improvements have been made in EIT, with the development of several indices to better assess the regional ventilation distribution. Interestingly, EIT is the only noninvasive bedside tool to monitor recruitment and risks of overdistension, which show not only the global effect but the regional changes

after an intervention. Because any ventilatory setting is always a compromise between harms and benefits for the lungs, EIT provides a wealth of valuable information on regional lung mechanics and ventilation. Some recent promising clinical studies suggest that EIT might work as a unique bedside tool to identify different phenotypes of ARDS, with important implications for PEEP or proning strategies (36, 78).

There are, however, many uncertainties regarding the EIT indices' normal values that deserve further investigation. Different EIT indices could be used for the same objectives, and the performance of these parameters should be compared with each other. Even if overdistension and collapse evaluation are frequently used for PEEP titration, this does not necessarily mean that this PEEP level will be the one that best mitigates ventilator-induced lung injury (79). In addition, the consequences and physiological impact of new recruitment indices such as the recruitment-to-inflation ratio should be better defined using EIT (63, 80). Importantly, evidence of EIT-guided ventilation leading to better outcomes is still in its infancy despite some promising initial steps (26, 81). Further patient-centered outcome research on EIT in patients with ARDS is highly warranted.

Routinely merging lung aeration and perfusion is also one of the next challenges of EIT devices. The evolution of EIT could raise hope for a direct estimation of stroke volume variation, which could offer completely noninvasive monitoring of heart–lung interactions (82). Moreover, EIT should be combined and synchronized with other respiratory signals (e.g., esophageal pressure). By doing so, clinicians could have access to several complementary and simultaneous pieces of information that have been unobtainable so far. Finally, if EIT is mainly used for patients with hypoxemic respiratory failure receiving invasive MV, expertise with its use could expand to other populations such as nonintubated patients and perioperative monitoring during abdominal or thoracic surgery.

Conclusions

Scientific research during the previous two decades has highlighted the ability of EIT to

evaluate the physiopathological mechanisms that cause hypoxemic respiratory failure and monitor MV settings. The simplicity of its use, without invasive procedures, combined with relevant physiological information provided at the bedside, led to growing interest from

clinicians. In addition, pulmonary perfusion analysis coupled with regional ventilation offers new possibilities for monitoring patients in critical condition. However, evidence of the impact of EIT-guided strategies on patient outcomes is still lacking. Numerous EIT

indices still deserve further investigation to understand their potential roles in the management of these patients. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

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